MSE 445 Materials Engineering Systems Analysis Fall 2025

Instructor:

Teaching Assistant:

Lecture:

Office Hours: After the first week of class I will set up regular office hours that will be conducted using Zoom. You are also welcome to drop by in person. Feel free to contact me by email at any time, and I am happy to set up meetings outside of office hours. _____ office hours will also be set up after the first week of class.

Websites: Brightspace will be used to post course information, assignments, class readings and any other relevant information.

Textbooks: 1) *Materials Selection in Mechanical Design*, 4th Edition, M. F. Ashby, Elsevier (2011), ISBN 1856176630 (available online through Purdue Libraries).
If you prefer a physical textbook I suggest purchasing the 5th edition of this book. Aside from a reordering of the chapters, most of the material is the same and the cost can be lower than the 4th edition since copies of the 4th edition are relatively scarce.
2) Materials and the Environment, 2nd edition, M.F. Ashby, Elsevier (2013), ISBN 978-0-12-385971-6 (available online through Purdue Libraries). Only a small section of this book will be used so I suggest you access the online version.

Description and Approach: MSE 445 introduces the practice of specifying materials and materials processing methods for commercial design applications. Topics will include: principles of materials and process selection, incorporation of environmental considerations, time value of money principles, material and process cost analysis, and the concept of statistical process control.

MSE 445 has assigned Problems of the Week, up to eight quizzes, two miniprojects, and a comprehensive final exam. Sample problems with solutions will be available to assist with quiz preparation. We will make extensive use of Granta Edupack, a software package that facilitates materials and process selection, both for the Problems of the Week and the mini-projects. Granta Edupack may be accessed at no cost using your computer, details of how to access the software will be provided early in the semester.

Homework and Problems of the Week: We will use Brightspace to distribute sample problems with solutions to help with quiz preparation approximately every other week. One or two additional "Problem(s) of the Week" will be assigned without solutions, and will be submitted for grading using Brightspace.

Quizzes and Final Exam: There will be quizzes approximately every other week. All quizzes during the semester are open book, and will be conducted using Brightspace. This means that you can take the quiz anywhere during our 50 minute class session. _____ will also be available if you are on campus. Excused absences from quizzes require verifiable documentation of the cause of the missed quiz (e.g., travel for job interview, doctor's note in case of illness). Unexcused absences will result in a zero score for the quiz.

We will have an in person open book comprehensive final exam. Details will be provided in the second half of the semester.

Quiz Preparation Advice: Since the quizzes and the final exam are open book there is a temptation to think that you will have time to look through notes and solved problems during the quiz. You will not have the time. This is by design. I want to know what you have learned and understand before you take the quiz, not what you can learn in real time during the quiz. I suggest that you prepare for each quiz the same as you prepare for a closed book quiz, plus some extra organization. For example, put together a summary sheet including any equations you think you might need, and maybe even some "highlights" or important concepts that you might have recognized while doing the practice problems. The process of making a summary sheet will help you study, and you won't need to spend time looking for things during the quiz. Further, you can collect your sheets for use during the final exam. Open book may give you the sense that you have time to look things up, but really you don't have the time to do that unless you know exactly where you can find the information.

Grading:	Problems of the Week	15%
	Quizzes	35%
	Miniprojects	20%
	Comprehensive Final Exam	30%

I will provide at least one "midterm update" during the semester that you may use to determine your approximate grade at that point in the semester.

Final grade assignments: The B/B- division will be determined by the median score.

Regrade requests: If you have concerns over the grading of an assignment or a quiz (anything from mistakes in adding scores to not understanding why you did not receive full credit for a question), please write down your concerns in an email message and send it to me with a scanned copy of your document. **Regrade requests must be received within two weeks after you have received your graded assignment or quiz** (e.g., I will not accept regrade requests for the first quiz a week before the final exam).

Academic Dishonesty Policies and Procedures:
Purdue University Regulations, Part 5, Section III-B-2-a describes the formal policies governing academic dishonesty. Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." More information related to academic integrity is available from the Office of the Dean of Students at:

https://www.purdue.edu/odos/osrr/academic-integrity/index.html

There are many dishonest ways to gain an advantage over another student in an assignment. The goal is not to list them here, but these rules cover any assignment for which the instructor will assign a grade (homework, quizzes, exams, laboratory reports, term paper, etc.). Rather, students should ask themselves this question when working on all class assignments: *"If I use this information, will the completed assignment represent only my efforts?"* If the answer is no, then don't do it. The test is simple. For example, turning in a term paper obtained from a website does not represent your efforts. Turning in copied homework from another student or a solutions manual does not represent your efforts either.

Consequences of Academic Dishonesty in MSE 445
The teaching staff for this course will diligently monitor academic dishonesty in all assignments. Students found to engage in academic dishonesty are subject to discipline beginning with a grade of zero for the assignment. Depending on the severity of the offense, other possible outcomes include a grade of F for the course, and a permanent letter added to your file. Serious or repeat incidents will be reported the Dean of Students for further action.

You can report issues of academic integrity that you observe, and may do so anonymously, through the Office of Student Rights and Responsibilities by calling 765-494-1250 or emailing osrr@purdue.edu.

Emergency Policies and Procedures:
In the event of a major campus emergency*, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. These changes would be posted on Brightspace*. In case of an extended disruption in which classes on campus are suspended the course will continue to the extent possible via *Brightspace*.